

Alberto Ferrari

Software Engineer | M.Sc. Student

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Web Portfolio

Work Experience

React Frontend Developer **Aug 2026 – Oct 2026**
Volio App Srl *Remote, Italy*
Developed reactive web interfaces and modular UI components with React.js and TypeScript. Managed client-side state, integrated AWS Cognito synchronization for user authentication, and optimized application runtime performance.

Student Academic Collaboration **May 2026 – Jul 2026**
BICF Library (Computer Science & Physics) – UniMi *Milan, Italy*
Provided operational support and technical assistance to students accessing university library resources and study rooms. Assisted in front-office library services and bibliographic cataloging.

Enterprise IT Systems Specialist **Mar 2025 – Sep 2025**
Emisfera *Verbania, Italy*
Full-time operational management of enterprise IT systems and platforms. Customization of Microsoft SharePoint environments, configuration and maintenance of the osTicket helpdesk system, and execution of database migrations.

Full Stack Developer **Jul 2024 – Mar 2025**
Marcatech *Varese, Italy*
Core contributor to the One-Staff enterprise web application for hotel staff administration. Engineered responsive React and TypeScript interfaces, developed scalable Node.js/NestJS backend services with PostgreSQL, and integrated robust REST APIs.

Education & Training

M.Sc. in Computer Science **Sep 2025 – Present**
University of Milan (UniMi) *Milan, Italy*
Curriculum focused on complex algorithms, network infrastructures, and cybersecurity.

B.Sc. in Computer Science **Sep 2021 – Dec 2024**
University of Insubria *Varese, Italy*
Final Grade: 99/110. Rigorous foundation in software engineering, relational databases, computer networks, and operating systems.

CyberChallenge.IT – Cybersecurity Training & CTF Competition **2024**
Cybersecurity National Lab (CINI) & University of Insubria *Italy*
Advanced training in applied cryptography, web application security, software defense, vulnerability assessment, networks, and CTF competition.

High School Diploma in Applied Sciences **Sep 2016 – Jun 2021**
IIS Lorenzo Cobianchi *Verbania, Italy*
Final Grade: 85/100. Strong STEM background in mathematics, physics, and computer science foundations.

Projects

Flight Monitor | *Apache Kafka, Cloud Computing, Distributed Systems* *github.com/Albitrix/flight-monitor*
Graduate project for the Cloud Computing M.Sc. course (UniMi). Designed and built a distributed system for real-time air traffic monitoring powered by Apache Kafka, architected to guarantee high availability (HA), fault tolerance, secure communication, and authentication.

New Gen Data Models | *MongoDB, NoSQL, Benchmark & Query Optimization* *github.com/Albitrix/new-gen-data-models*
Graduate project for the New Generation Data Models and DBMSs M.Sc. course (UniMi). Designed and developed a transactional MongoDB database architecture, featuring index optimization and benchmarking across complex analytical query workloads.

Technical & Language Skills

Programming Languages: TypeScript, JavaScript, Python, Java, C
Web & Backend Development: React.js, Node.js, NestJS, Next.js, Apache Kafka, REST APIs, HTML, HTTP/HTTPS
Databases & Systems: SQL, PostgreSQL, MongoDB, Neo4j, NoSQL, Linux, Docker, Git, SharePoint, osTicket
Cybersecurity: Web Security, Applied Cryptography, Software Defense, Vulnerability Assessment
Languages: Italian (Native), English (Fluent)